

CARBOXYLATED NBR: PATENT LANDSCAPE AND SYNTHESIS/USE ANALYSIS

1. ABSTRACT

This report reviews patent evidence relevant to carboxylated nitrile butadiene rubber (NBR), with emphasis on how acid- or acid anhydride-modified acrylonitrile-butadiene copolymers are used in heat-activable adhesive systems. The primary patent in the manifest does not disclose a standalone polymerization recipe for making the carboxylated NBR; instead, it focuses on formulation and end-use performance of an adhesive tape for flexible printed circuit boards (FPCBs). The target material is defined as an acid- or acid anhydride-modified acrylonitrile-butadiene copolymer, used with epoxy resin at a weight ratio greater than 1.5 and without an additional nonpolymer hardener. The evidence indicates that the carboxylated nitrile rubber functions as both elastomer and reactive component, enabling heat-induced crosslinking with epoxy resin while maintaining low flow at elevated temperature.

Across the disclosed examples, the patent compares a carboxylated NBR/epoxy formulation against a formulation containing unmodified nitrile rubber and a hardener, and against a formulation with added polyterpene resin. The carboxylated NBR examples show improved bond strength and temperature stability, while all examples pass solder-bath resistance. The patent also specifies that suitable nitrile rubbers may have acrylonitrile contents of 15% to 50% by weight, and that the adhesive crosslinks above 150°C. The synthesis/process emphasis is therefore on solution coating of adhesive compositions, drying, lamination at 100°C, and final bonding at 200°C under 1.3 MPa for one hour, rather than on polymer synthesis.

Overall, the patent landscape here is formulation-driven: carboxylated NBR is treated as a reactive elastomer for high-temperature adhesive tapes, not as a separately synthesized polymer in the examples. The most important process trend is the use of acid functionality to eliminate the need for external epoxy hardener while preserving flexibility, solvent processability, and thermal resistance in FPCB bonding applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20100147462A1
- Patent Title: ... strip that can be activated by heat and is based on carboxylated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a heat-activable adhesive tape formulation based on an acid- or acid anhydride-modified acrylonitrile-butadiene copolymer combined with an epoxy resin. The carboxylated nitrile rubber is dissolved in butanone, coated onto release paper, dried, and later laminated/bonded under heat and pressure to form a crosslinked adhesive joint. - Nitrile rubber composition: 26% by weight acrylonitrile — Example 1 - Acid modification: 7% by weight — Example 1 - Epoxy resin epoxide equivalent: 184 — Example 1 - Coating thickness: 25 μm — Example 1 - Polyterpene resin addition: 20 parts by weight — Example 2 - Breon N41H80 acrylonitrile content: 41% by weight — Example 3 - Breon N41H80 Mooney viscosity ML 1+4 at 100°C: 72 to 88 — Example 3 - Dyhard 100-S addition: 5 parts by weight — Example 4 - Lamination temperature: 100°C — Bonding of FPCBs - Press temperature: 200°C — Bonding of FPCBs - Press pressure: 1.3 MPa — Bonding of FPCBs - Press time: 1 hour — Bonding of FPCBs - T-peel test angle: 180° — Test Methods - T-peel test rate: 50 mm/min — Test Methods - T-peel test temperature: 20°C — Test Methods - Relative humidity: 50% — Test Methods - Static peel test temperature: 70°C — Temperature Stability - Static peel load: 500 g — Temperature Stability - Solder bath temperature: 288°C — Solder Bath Resistance - Solder bath exposure time: 10 seconds — Solder Bath Resistance - Weight ratio of components a/b: greater than 1.5 — Claims - Acrylonitrile content range: 15% to 50% by weight — Claims - Crosslinking temperature: above 150°C — Claims

Relevant Experimental Evidence - Example 1: Carboxylated NBR with epoxy resin produced a 25 μm adhesive layer and served as the main inventive formulation. - Example 2: Addition of 20 parts by weight polyterpene resin to the Example 1 formulation was evaluated as a modified adhesive composition. - Example 3: Comparative unmodified nitrile rubber formulation showed very low bond strength without crosslinking. - Example 4: Comparative formulation with dicyandiamide hardener was tested alongside the inventive system and used for bonding evaluation.

Technical Relevance This patent is directly relevant because it uses acid-modified nitrile rubber as the reactive elastomeric component in a carboxylated NBR-based adhesive system. It demonstrates the functional value of carboxylation in enabling epoxy crosslinking without an added nonpolymer hardener, which is central to high-temperature NBR applications.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The evidence supports carboxylated NBR as a reactive, formulation-enabling elastomer rather than a separately detailed synthesis target in this patent set. The key disclosed design space is an acid- or acid anhydride-modified acrylonitrile-butadiene copolymer with 15% to 50% acrylonitrile, combined with epoxy resin at a weight ratio greater than 1.5 and processed from solution into heat-activable adhesive tapes. For practical synthesis/formulation, the most defensible parameters from the record are acid-modified NBR, epoxy resin, butanone solution coating, drying to a 25 µm layer in Example 1, lamination at 100°C, final bonding at 200°C and 1.3 MPa for 1 hour, and crosslinking above 150°C. The patent indicates that carboxylation improves bond strength and temperature stability while eliminating the need for an external epoxy hardener.

5. REFERENCES

- US20100147462A1 | ... strip that can be activated by heat and is based on carboxylated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US20100147462A1/en>